1) a)

$y^2 = (x+1)^2 = x^2 + 2x + 1$

1) b)

$y^2 = (3x-1)^2 = 9x^2 - 6x + 1$

1) c)

$y^2 = \left(x - \frac{1}{x}\right)^2 = x^2 - 2 + \frac{1}{x^2}$

1) d)

$y^2 = (x^2+5)^2 = x^4 + 10x^2 + 25$

1) e)

$y^2 = \left(2\sqrt{x} + 3\right)^2 = 4x + 12\sqrt{x} + 9$

1) f)

$y^2 = \left(\frac{1}{\sqrt{x^3}} + 1\right)^2 = \frac{1}{x^3} + \frac{2}{\sqrt{x^3}} + 1$

2) a)

$\pi \int_1^5 y^2 \, \mathrm{d}x = \pi \int_1^5 (x-1)^2 \, \mathrm{d}x = \pi \int_1^5 (x^2 - 2x + 1) \, \mathrm{d}x$

$= \pi \left[\frac{1}{3}x^3 - x^2 + x\right]_1^5$

$= \pi \left[\left(\frac{5^3}{3} - 5^2 + 5\right) - \left(\frac{1^3}{3} - 1^2 + 1\right)\right]$

$= \frac{64}{3}\pi$ units^3

2) b)

$$\pi\int_0^2 y^2\,\mathrm{d}x = \pi\int_0^2 (x^2+2)^2\,\mathrm{d}x = \pi\int_0^2 (x^4+4x^2+4)\,\mathrm{d}x$$
$$= \pi\left[\frac{1}{5}x^5+\frac{4}{3}x^3+4x\right]_0^2$$
$$= \pi\left[\left(\frac{2^5}{5}+\frac{4\times 2^3}{3}+8\right)-(0)\right]$$
$$= \frac{376}{15}\pi \text{ units}^3$$

2) c)

$$\pi\int_1^3 y^2\,\mathrm{d}x = \pi\int_1^3 \left(x^{-\frac{3}{2}}\right)^2\,\mathrm{d}x = \pi\int_1^3 x^{-3}\,\mathrm{d}x$$
$$= \pi\left[-\frac{1}{2}x^{-2}\right]_1^3$$
$$= \pi\left[\left(-\frac{1}{2}\times 3^{-2}\right)-\left(-\frac{1}{2}\times 1^{-2}\right)\right]$$
$$= \frac{4}{9}\pi \text{ units}^3$$

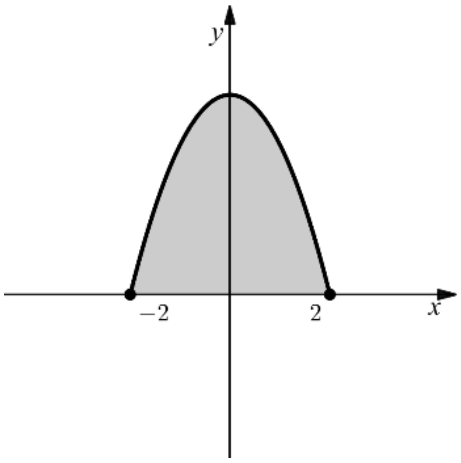
2) d)

$$\pi\int_1^4 y^2\,\mathrm{d}x = \pi\int_1^4 \left(2+\sqrt{x}\right)^2\,\mathrm{d}x = \pi\int_1^4 \left(4+4\sqrt{x}+x\right)\mathrm{d}x$$
$$= \pi\left[4x+\frac{8}{3}x^{\frac{3}{2}}+\frac{1}{2}x^2\right]_1^4$$
$$= \pi\left[\left(4\times 4+\frac{8}{3}\times 4^{\frac{3}{2}}+\frac{1}{2}\times 4^2\right)-\left(4+\frac{8}{3}+\frac{1}{2}\right)\right]$$
$$= \frac{229}{6}\pi \text{ units}^3$$

3)

$$\pi\int_1^2 y^2\,\mathrm{d}x = \pi\int_1^2 (x^3+2)^2\,\mathrm{d}x = \pi\int_1^2 (x^6+4x^3+4)\,\mathrm{d}x$$
$$= \pi\left[\frac{1}{7}x^7+x^4+4x\right]_1^2$$
$$= \pi\left[\left(\frac{2^7}{7}+2^4+8\right)-\left(\frac{1^7}{7}+1^4+4\right)\right]$$
$$= \frac{260}{7}\pi$$
$$= 116.7 \text{ cm}^3$$

4) a)



4) b)

$$\pi \int_{-2}^{2} y^2 \, \mathrm{d}x = \pi \int_{-2}^{2} (4 - x^2)^2 \, \mathrm{d}x = \pi \int_{-2}^{2} (16 - 8x^2 + x^4) \, \mathrm{d}x$$

$$= \pi \left[16x - \frac{8}{3}x^3 + \frac{1}{5}x^5 \right]_{-2}^{2}$$

$$= \pi \left[\left(32 - \frac{8}{3} \times 2^3 + \frac{1}{5} \times 2^5 \right) - \left(-32 - \frac{8}{3} \times (-2)^3 + \frac{1}{5} \times (-2)^5 \right) \right]$$

$$= \frac{512}{15}\pi \text{ units}^3$$

5)

$$\pi \int_{0}^{h} r^2 \, \mathrm{d}x$$

$$= \pi [r^2 x]_0^h$$

$$= \pi [(r^2 h) - (0)]$$

$$= \pi r^2 h \text{ as req.}$$

6) a)

$y = kx$ and $x = h$ intersect when $y = r$

$\therefore k = \frac{r}{h}$

6) b)

$$\pi\int_0^h y^2\,\mathrm{d}x = \pi\int_0^h \left(\frac{r}{h}x\right)^2\mathrm{d}x = \pi\int_0^h \frac{r^2}{h^2}x^2\,\mathrm{d}x$$
$$= \pi\left[\frac{r^2}{3h^2}x^3\right]_0^h$$
$$= \pi\left[\left(\frac{r^2}{3h^2}h^3\right) - (0)\right]$$
$$= \frac{1}{3}\pi r^2 h \text{ as req.}$$

7) a)

$$\pi\int_1^3 y^2\,\mathrm{d}x = \pi\int_1^3 \left(\sqrt{x-1}\right)^2\mathrm{d}x = \pi\int_1^3 (x-1)\,\mathrm{d}x$$
$$= \pi\left[\frac{1}{2}x^2 - x\right]_1^3$$
$$= \pi\left[\left(\frac{9}{2}-3\right) - \left(\frac{1}{2}-1\right)\right]$$
$$= 2\pi \text{ units}^3$$

7) b)

$$\pi\int_{-1}^8 y^2\,\mathrm{d}x = \pi\int_{-1}^8 \left(2+x^{\frac{1}{3}}\right)^2\mathrm{d}x = \pi\int_{-1}^8 \left(4+4x^{\frac{1}{3}}+x^{\frac{2}{3}}\right)\mathrm{d}x$$
$$= \pi\left[4x+3x^{\frac{4}{3}}+\frac{3}{5}x^{\frac{5}{3}}\right]_{-1}^8$$
$$= \pi\left[\left(32+48+\frac{96}{5}\right) - \left(-4+3-\frac{3}{5}\right)\right]$$
$$= \frac{504}{5}\pi \text{ units}^3$$

7) c)

$-2x^2+6x = 2x(3-x)$

The curve meets the axis when $x = 0$ and $x = 3$

$$\pi\int_0^3 y^2\,\mathrm{d}x = \pi\int_0^3 (-2x^2+6x)^2\,\mathrm{d}x = \pi\int_0^3 (4x^4-24x^3+36x^2)\,\mathrm{d}x$$
$$= \pi\left[\frac{4}{5}x^5 - 6x^4 + 12x^3\right]_0^3$$
$$= \pi\left[\left(\frac{972}{5} - 486 + 324\right) - (0)\right]$$
$$= \frac{162}{5}\pi \text{ units}^3$$

8)

$x^3 - 4x^2 = x^2(x-4)$

The curve meets the x-axis when $x = 0$ and $x = 4$

$$\pi \int_0^4 y^2 \, dx = \pi \int_0^4 (x^3 - 4x^2)^2 \, dx = \pi \int_0^4 (x^6 - 8x^5 + 16x^4) \, dx$$

$$= \pi \left[\frac{1}{7}x^7 - \frac{4}{3}x^6 + \frac{16}{5}x^5 \right]_0^4$$

$$= \pi \left[\left(\frac{16384}{7} - \frac{16384}{3} + \frac{16384}{5} \right) - (0) \right]$$

$$= \frac{16384}{105}\pi$$

$= 490.2$ units^3

9) a)

$y = 1 + \frac{1}{1} = 2$ and $y = 1.5 + \frac{1}{1.5} = \frac{13}{6}$

So radii are 2 and $\frac{13}{6}$

Volume is $0.5 \times \pi \times 2^2 + 0.5 \times \pi \times \left(\frac{13}{6}\right)^2 = 13.7$ units^3 (3 SF)

9) b)

He could increase the number of cylinders he is using.

9) c)

$$\pi \int_{0.5}^{1.5} y^2 \, dx = \pi \int_{0.5}^{1.5} \left(x + \frac{1}{x} \right)^2 dx = \pi \int_{0.5}^{1.5} (x^2 + 2 + x^{-2}) \, dx$$

$$= \pi \left[\frac{1}{3}x^3 + 2x - x^{-1} \right]_{0.5}^{1.5}$$

$$= \pi \left[\left(\frac{1.5^3}{3} + 3 - 1.5^{-1} \right) - \left(\frac{0.5^3}{3} + 1 - 0.5^{-1} \right) \right]$$

$$= \frac{53}{12}\pi \text{ units}^3$$

10) a)

$\pi \int_0^r y^2 \, \mathrm{d}x = \pi \int_0^r (r^2 - x^2) \, \mathrm{d}x$

$= \pi \left[r^2 x - \frac{1}{3} x^3 \right]_0^r$

$= \pi \left[\left(r^3 - \frac{1}{3} r^3 \right) - (0) \right]$

$= \frac{2}{3} \pi r^3$ units^3

10) b)

$2 \times \frac{2}{3} \pi r^3 = \frac{4}{3} \pi r^3$ as req.

11) a)

When $y = 0$:

$-6x = 9 - x^3$

$x^3 - 6x - 9 = 0$

Let $\mathrm{f}(x) = x^3 - 6x - 9$

$\mathrm{f}(3) = 27 - 18 - 9 = 0$ so $(x - 3)$ is a factor so $x = 3$ is a solution.

The curve meets the x-axis when $x = 3$

11) b)

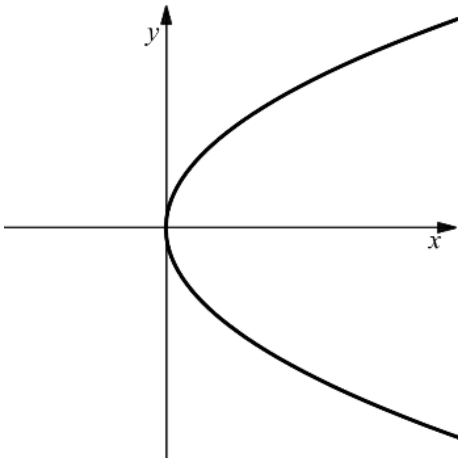
$\pi \int_{-3}^3 y^2 \, \mathrm{d}x = \pi \int_{-3}^3 \left(\frac{9}{2} - \frac{1}{2} x^3 + 3x \right) \mathrm{d}x$

$= \pi \left[\frac{9}{2} x - \frac{1}{8} x^4 + \frac{3}{2} x^2 \right]_{-3}^3$

$= \pi \left[\left(\frac{27}{2} - \frac{81}{8} + \frac{27}{2} \right) - \left(-\frac{27}{2} - \frac{81}{8} + \frac{27}{2} \right) \right]$

$= 27\pi$ units^3

12) a)



12) b)

$\pi \int_0^d y^2 \, \mathrm{d}x = \pi \int_0^d 3x \, \mathrm{d}x$

$= \pi \left[\frac{3}{2}x^2\right]_0^d$

$= \pi \left[\left(\frac{3}{2}d^2\right) - 0\right]$

$= \frac{3}{2}\pi d^2$

$500 = \frac{3}{2}\pi d^2$ so $d = \sqrt{\frac{1000}{3\pi}} = 10.3$ cm

12) c)

That the material the bowl is has no thickness

13) a)

$180 = \pi \int_0^6 y^2 \, \mathrm{d}x = \pi \int_0^6 kx \, \mathrm{d}x$

$= \pi \left[\frac{k}{2}x^2\right]_0^6$

$= \pi[(18k) - 0]$

$= 13\pi k$

$180 = 36\pi k$ so $k = \frac{10}{\pi}$

So the model is $y^2 = \frac{10}{\pi}x$

The radius occurs when $x = 6$ so $r = \sqrt{\frac{10}{\pi} \times 6} = 4.4$ cm

13) b)

Reduce the value of k and increase the height of the glass using the formula

14)

$$\pi\int_1^{\sqrt{3}} y^2\,\mathrm{d}x = \pi\int_1^{\sqrt{3}} (-x^2+3)^2\,\mathrm{d}x = \pi\int_1^{\sqrt{3}} x^4-6x^2+9\,\mathrm{d}x$$

$$= \pi\left[\frac{1}{5}x^5 - 2x^3 + 9x\right]_1^{\sqrt{3}}$$

$$= \pi\left[\left(\frac{1}{5}\times\sqrt{3}^5 - 2\times\sqrt{3}^3 + 9\sqrt{3}\right) - \left(\frac{1}{5}-2+9\right)\right]$$

$$= \pi\left[\left(\frac{9}{5}\sqrt{3} - 6\sqrt{3} + 9\sqrt{3}\right) - \left(\frac{1}{5}-2+9\right)\right]$$

$$= \left(\frac{24}{5}\sqrt{3} - \frac{36}{5}\right)\pi$$

Volume of cone: $\frac{1}{3}\pi\times 2^2\times 1 = \frac{4}{3}\pi$

Total volume: $\frac{4}{3}\pi + \left(\frac{24}{5}\sqrt{3} - \frac{36}{5}\right)\pi = \left(\frac{24}{5}\sqrt{3} - \frac{88}{15}\right)\pi$ units^3

15) a)

$y = 8-3x$ so $A\left(\frac{8}{3},0\right)$

$\frac{1}{4}x^3 = 8-3x \Rightarrow x^3 = 32-12x \Rightarrow x^3+12x-32=0$

Let $\mathrm{f}(x) = x^3+12x-32$

$\mathrm{f}(2) = 8+24-32 = 0$ so $(x-2)$ is a factor so $x=2$ is a solution so $B(2,2)$

$A\left(\frac{8}{3},0\right)$ and $B(2,2)$

15) b)

$$\pi\int_0^2 y^2\,\mathrm{d}x = \pi\int_0^2 \frac{1}{16}x^6\,\mathrm{d}x$$

$$= \pi\left[\frac{1}{112}x^7\right]_0^2$$

$$= \pi\left[\frac{8}{7}-0\right]$$

$$= \frac{8}{7}\pi$$

Volume of cone: $\frac{1}{3}\pi\times 2^2\times\left(\frac{8}{3}-2\right) = \frac{8}{9}\pi$

Total volume: $\frac{8}{7}\pi + \frac{8}{9}\pi = \frac{128}{63}\pi$ units^3

16) a)

$2x + 0.5x^2 = 6$

$0.5x^2 + 2x - 6 = 0$

Gives $x = 2$ or $x = -6$ so $x = 2$

$(2, 2)$

16) b)

$$\pi \int_2^3 y^2 \, \mathrm{d}x = \pi \int_2^3 \frac{1}{4}x^4 \, \mathrm{d}x$$

$$= \pi \left[\frac{1}{20}x^5\right]_2^3$$

$$= \pi \left[\frac{243}{20} - \frac{8}{5}\right]$$

$$= \frac{211}{20}\pi$$

Volume of cone: $\frac{1}{3}\pi \times 2^2 \times (3-2) = \frac{4}{3}\pi$

Total volume: $\frac{211}{20}\pi - \frac{4}{3}\pi = \frac{553}{60}\pi$ units^3

17) a)

$$\frac{1}{4} \times \pi \int_1^4 y^2 \, \mathrm{d}x = \frac{\pi}{4} \int_1^4 \left(\frac{1}{4}x^2 + x + 1\right) \mathrm{d}x$$

$$= \frac{\pi}{4}\left[\frac{1}{12}x^3 + \frac{1}{2}x^2 + x\right]_1^4$$

$$= \frac{\pi}{4}\left[\left(\frac{16}{3} + 8 + 4\right) - \left(\frac{1}{12} + \frac{1}{2} + 1\right)\right]$$

$$= \frac{63}{16}\pi \text{ units}^3$$

17) b)

$$\frac{1}{2} \times \pi \int_1^2 y^2 \, \mathrm{d}x = \frac{\pi}{2} \int_1^2 (x^4 - 2x^2 + 1) \, \mathrm{d}x$$

$$= \frac{\pi}{2}\left[\frac{1}{5}x^5 - \frac{2}{3}x^3 + x\right]_1^2$$

$$= \frac{\pi}{2}\left[\left(\frac{32}{5} - \frac{16}{3} + 2\right) - \left(\frac{1}{5} - \frac{2}{3} + 1\right)\right]$$

$$= \frac{19}{15}\pi \text{ units}^3$$

17) c)

$$\frac{1}{4}\times\pi\int_0^1 y^2\,\mathrm{d}x=\frac{\pi}{4}\int_0^1\left(16-8\sqrt{x}+x\right)\mathrm{d}x$$

$$=\frac{\pi}{4}\left[16x-\frac{16}{3}x^{\frac{3}{2}}+\frac{1}{2}x^2\right]_0^3$$

$$=\frac{\pi}{4}\left[\left(16-\frac{16}{3}+\frac{1}{2}\right)-(0)\right]$$

$$=\frac{67}{24}\pi \text{ units}^3$$

18)

$$\pi\int_{-4}^4 y^2\,\mathrm{d}x=\pi\int_{-4}^4(16-x^2)\,\mathrm{d}x$$

$$=\pi\left[16x-\frac{1}{3}x^3\right]_{-4}^4$$

$$=\pi\left[\left(64-\frac{64}{3}\right)-\left(-64+\frac{64}{3}\right)\right]$$

$$=\frac{256}{3}\pi$$

and

$$\pi\int_{-2}^2 y^2\,\mathrm{d}x=\pi\int_{-2}^2 2^2\,\mathrm{d}x$$

$$=\pi[4x]_{-2}^2$$

$$=\pi[8-(-8)]$$

$$=16\pi$$

Total volume: $\frac{256}{3}\pi-16\pi=\frac{208}{3}\pi$ units^3

19) a)

$2x=-x^2+4x\Rightarrow x^2-4x+2x=0\Rightarrow x^2-2x=0\Rightarrow x(x-2)=0$

So $x=2$

$A(2,4)$

19) b)

$B(2,-4)$

19) c)

Volume of cone: $\frac{1}{3}\pi \times 4^2 \times 2 = \frac{32}{3}\pi$

and

$$\pi \int_2^4 y^2 \, \mathrm{d}x = \pi \int_2^4 (-x^2 + 4x)^2 \, \mathrm{d}x = \pi \int_2^4 (x^4 - 8x^3 + 16x^2) \, \mathrm{d}x$$

$$= \pi \left[\frac{1}{5}x^5 - 2x^4 + \frac{16}{3}x^3\right]_2^4$$

$$= \pi \left[\left(\frac{1024}{5} - 512 + \frac{1024}{3}\right) - \left(\frac{32}{5} - 32 + \frac{128}{3}\right)\right]$$

$$= \frac{256}{15}\pi$$

Total volume: $\frac{32}{3}\pi + \frac{256}{15}\pi = \frac{416}{15}\pi$ units^3

20)

$$\pi \int_{-3}^3 y^2 \, \mathrm{d}x = \pi \int_{-3}^3 (9 - x^2) \, \mathrm{d}x$$

$$= \pi \left[9x - \frac{1}{3}x^3\right]_{-3}^3$$

$$= \pi[(27 - 9) - (-27 + 9)]$$

$$= 36\pi$$

and

$$\pi \int_{-2}^2 y^2 \, \mathrm{d}x = \pi \int_{-2}^2 (4 - x^2) \, \mathrm{d}x$$

$$= \pi \left[4x - \frac{1}{3}x^3\right]_{-2}^2$$

$$= \pi \left[\left(8 - \frac{8}{3}\right) - \left(-8 + \frac{8}{3}\right)\right]$$

$$= \frac{32}{3}\pi$$

Total volume: $36\pi - \frac{32}{3}\pi = \frac{76}{3}\pi$ units^3

21) a)

$0 = -\frac{1}{8}x^2 + 2 \Rightarrow x = 4$

$A(4, 0)$

21) b)

$$\pi \int_0^4 y^2 \, \mathrm{d}x = \pi \int_0^4 (16 - x^2) \, \mathrm{d}x$$

$$= \pi \left[16x - \frac{1}{3}x^3\right]_0^4$$

$$= \pi \left[\left(64 - \frac{64}{3}\right) - (0)\right]$$

$$= \frac{128}{3}\pi$$

and

$$\pi \int_0^4 y^2 \, \mathrm{d}x = \pi \int_0^4 \left(-\frac{1}{8}x^2 + 2\right)^2 \mathrm{d}x = \pi \int_0^4 \left(\frac{1}{64}x^4 - \frac{1}{2}x^2 + 4\right) \mathrm{d}x$$

$$= \pi \left[\frac{1}{320}x^5 - \frac{1}{6}x^3 + 4x\right]_0^4$$

$$= \pi \left[\left(\frac{1}{320} \times 4^5 - \frac{1}{6} \times 4^3 + 16\right) - (0)\right]$$

$$= \frac{128}{15}\pi$$

Total volume: $\frac{128}{3}\pi - \frac{128}{15}\pi = \frac{512}{15}\pi$ units^3

22) a)

$$\pi \int_1^5 x^2 \, dy = \pi \int_1^5 (y^2 + 4y + 4) \, dy$$
$$= \pi \left[\frac{1}{3} y^3 + 2y^2 + 4y \right]_1^5$$
$$= \pi \left[\left(\frac{125}{3} + 50 + 20 \right) - \left(\frac{1}{3} + 2 + 4 \right) \right]$$
$$= \frac{316}{3} \pi \text{ units}^3$$

22) b)

$$\pi \int_0^1 x^2 \, dy = \pi \int_0^1 y^6 \, dy$$
$$= \pi \left[\frac{1}{7} y^7 \right]_0^1$$
$$= \pi \left[\frac{1}{7} - 0 \right]$$
$$= \frac{1}{7} \pi \text{ units}^3$$

22) c)

$$\pi \int_0^4 x^2 \, dy = \pi \int_0^4 y \, dy$$
$$= \pi \left[\frac{1}{2} y^2 \right]_0^4$$
$$= \pi [8 - 0]$$
$$= 8\pi \text{ units}^3$$

23)

$$\pi \int_4^{10} x^2 \, dy = \pi \int_4^{10} (y + 6) \, dy$$
$$= \pi \left[\frac{1}{2} y^2 + 6y \right]_4^{10}$$
$$= \pi \left[\left(\frac{1}{2} \times 10^2 + 60 \right) - \left(\frac{1}{2} \times 4^2 + 24 \right) \right]$$
$$= 78\pi \text{ as req.}$$

24)

$$\pi \int_0^4 x^2 \, \mathrm{d}y = \pi \int_0^4 \left(y - y^{\frac{1}{2}} + 1\right) \mathrm{d}y$$
$$= \pi \left[\frac{1}{2}y^2 - \frac{2}{3}y^{\frac{3}{2}} + y\right]_0^4$$
$$= \pi \left[\left(8 - \frac{16}{3} + 4\right) - (0)\right]$$
$$= \frac{20}{3}\pi \text{ units}^3$$

25)

$$\pi \int_1^4 x^2 \, \mathrm{d}y = \pi \int_1^4 (y^2 - 4y + 5)^2 \, \mathrm{d}y = \pi \int_1^4 (y^4 - 8y^3 + 26y^2 - 40y + 25) \, \mathrm{d}y$$
$$= \pi \left[\frac{1}{5}y^5 - 2y^4 + \frac{26}{3}y^3 - 20y^2 + 25y\right]_1^4$$
$$= \pi \left[\left(\frac{1024}{5} - 512 + \frac{1664}{3} - 320 + 100\right) - \left(\frac{1}{5} - 2 + \frac{26}{3} - 20 + 25\right)\right]$$
$$= \frac{78}{5}\pi \text{ units}^3$$

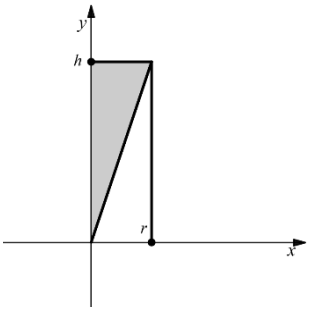
26)

$$\pi \int_0^1 x^2 \, \mathrm{d}y = \pi \int_0^1 (3 - y^3 - 2y) \, \mathrm{d}y$$
$$= \pi \left[3y - \frac{1}{4}y^4 - y^2\right]_0^1$$
$$= \pi \left[\left(3 - \frac{1}{4} - 1\right) - (0)\right]$$
$$= \frac{7}{4}\pi \text{ units}^3$$

27)

$$\pi \int_0^h x^2 \, \mathrm{d}y = \pi \int_0^h r^2 \, \mathrm{d}y$$
$$= \pi [r^2 y]_0^h$$
$$= \pi [(r^2 h) - (0)]$$
$$= \pi r^2 h \text{ units}^3$$

28)



$y = kx$ and $y = h$ intersect when $x = r$

$\therefore k = \frac{h}{r}$

$$\pi \int_0^h x^2 \, \mathrm{d}y = \pi \int_0^h \left(\frac{y}{k}\right)^2 \mathrm{d}y = \pi \int_0^h \frac{r^2}{h^2} y^2 \, \mathrm{d}y$$

$$= \pi \left[\frac{r^2}{3h^2} y^3\right]_0^h$$

$$= \pi \left[\left(\frac{r^2}{3h^2} h^3\right) - (0)\right]$$

$= \frac{1}{3}\pi r^2 h$ as req.

29) a)

$A(-1, 0)$, $B(1, 0)$ and $C(0, 1)$

29) b)

$$\pi \int_0^1 x^2 \, \mathrm{d}y = \pi \int_0^1 1 - y \, \mathrm{d}y$$

$$= \pi \left[y - \frac{1}{2} y^2\right]_0^1$$

$$= \pi \left[\left(1 - \frac{1}{2}\right) - (0)\right]$$

$= \frac{1}{2}\pi \approx 1.57$ m^3

29) c)

$1600 \times \frac{1}{2}\pi = 2510$ kg

29) d)

This will only be true if the sand is perfectly symmetrical and follows the curve exactly. There may be inconsistencies in the sand, such as varying compactness so that the density is not the same throughout.

30) a)

$V = \pi \int_0^d x^2 \, \mathrm{d}y$

$\pi \int_0^d \left(\frac{y}{0.4}\right) \mathrm{d}y$

$= \pi \left[\frac{y^2}{0.8}\right]_0^d$

$= \pi \left[\left(\frac{d^2}{0.8}\right) - 0\right]$

$= \frac{\pi}{0.8} d^2$

$\approx 3.93d^2$ as req.

30) b)

$3.93 \times 6^2 = 141$ cm^3 (3 SF)

30) c)

$360 = 3.93d^2$ gives $d \approx 9.57$ cm

31) a)

$V = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi \times 4^2 \times 6 = 32\pi \approx 101$ cm^3

31) b)

$\frac{4}{6}r = \frac{2}{3}h$ so $V = \frac{1}{3}\pi \left(\frac{2}{3}h\right)^2 h$

$\frac{32\pi}{2} = \frac{1}{3}\pi \left(\frac{2}{3}h\right)^2 h$

$48 = \frac{4}{9}h^3$

$h^3 = 108$

$h = 4.76$ cm

31) c)

Gradient: $\frac{6}{4} = \frac{3}{2}$ and goes through the y-axis at $(0, 8)$

$y = \frac{3}{2}x + 8 \Rightarrow y - 8 = \frac{3}{2}x$

$\therefore x = \frac{2y-16}{3}$

31) d)

$\pi \int_8^{14} x^2 \, \mathrm{d}y = \pi \int_8^{14} \left(\frac{2}{3}(y-8)\right)^2 \mathrm{d}y = \pi \int_8^{14} \left(\frac{4}{9}y^2 - \frac{64}{9}y + \frac{256}{9}\right) \mathrm{d}y$

$= \pi \left[\frac{4}{27}y^3 - \frac{32}{9}y^2 + \frac{256}{9}y\right]_8^{14}$

$= \pi \left[\left(\frac{4}{27} \times 14^3 - \frac{32}{9} \times 14^2 + \frac{256}{9} \times 14\right) - \left(\frac{4}{27} \times 8^3 - \frac{32}{9} \times 8^2 + \frac{256}{9} \times 8\right)\right]$

$= 32\pi \approx 101$ cm^3

32)

Volume of cone: $\frac{1}{3}\pi \times 2^2 \times (8-6) = \frac{8\pi}{3}$

$\pi \int_0^6 x^2 \, \mathrm{d}y = \pi \int_0^6 \frac{y}{1.5} \mathrm{d}y$

$= \pi \left[\frac{1}{3}y^2\right]_0^6$

$= \pi \left[\left(\frac{6^2}{3}\right) - (0)\right]$

$= 12\pi$

Total volume is $\frac{44\pi}{3}$ units^3

33) a)

$3.2 = k\sqrt{16} - 0.05 \times 16^2$

Gives $k = 4$

33) b)

Cylinder: $\pi \times 3.2^2 \times 2 = \frac{512}{25}\pi$

$\pi \int_0^{16} x^2 \, dy = \pi \int_0^{16} \left(4\sqrt{y} - 0.05y^2\right)^2 \, dy = \pi \int_0^{16} 16y - \frac{2}{5}y^{\frac{5}{2}} + \frac{1}{400}y^4 \, dy$

$= \pi \left[8y^2 - \frac{4}{35}y^{\frac{7}{2}} + \frac{1}{2000}y^5\right]_0^{16}$

$= \pi \left[\left(2048 - \frac{65536}{35} + \frac{65536}{125}\right) - (0)\right]$

$= \frac{612352}{875}\pi$

Volume: $\frac{512}{25}\pi + \frac{612352}{875}\pi = 2260$ cm^3 (3 SF)

33) c) i)

The curve may not represent the shape of the vase accurately, for example, the curve does not flatten out to give a surface for the vase to stand on.

33) c) ii)

The volume calculation does not allow for the thickness of the material that it is made out of.

34)

$y = x^2 - 4x + 4 = (x-2)^2$

So that $\sqrt{y} = x - 2$

$\therefore x = 2 + \sqrt{y}$ and so $x^2 = 4 + 4\sqrt{y} + y$

$\pi \int_0^3 x^2 \, dy = \pi \int_0^3 4 + 4\sqrt{y} + y \, dy$

$= \pi \left[4y + \frac{8}{3}y^{\frac{3}{2}} + \frac{1}{2}y^2\right]_0^3$

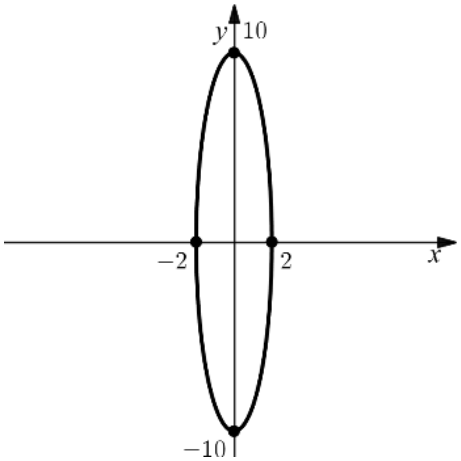
$= \pi \left[\left(12 + 8\sqrt{3} + \frac{9}{2}\right) - (0)\right]$

$= \left(\frac{33}{2} + 8\sqrt{3}\right)\pi$

35) a)

When $x = 0$, $\frac{y^2}{100} = 1$ so $y = \pm 10$

When $y = 0$, $\frac{x^2}{4} = 1$ so $y = \pm 2$



35) b)

$\frac{y^2}{100} = 1 - \frac{x^2}{4}$ so $y^2 = 100 - 25x^2$

$\pi \int_{-2}^{2} y^2 \, \mathrm{d}x = \pi \int_{-2}^{2} 100 - 25x^2 \, \mathrm{d}x$

$= \pi \left[100x - \frac{25}{3}x^3\right]_{-2}^{2}$

$= \pi \left[\left(200 - \frac{200}{3}\right) - \left(-200 + \frac{200}{3}\right)\right]$

$= \frac{800}{3}\pi$ units^3

35) c)

$\frac{x^2}{4} = 1 - \frac{y^2}{100}$ so $x^2 = 4 - \frac{1}{25}y^2$

$\pi \int_{-10}^{10} x^2 \, \mathrm{d}y = \pi \int_{-10}^{10} 4 - \frac{1}{25}y^2 \, \mathrm{d}y$

$= \pi \left[4y - \frac{1}{75}y^3\right]_{-10}^{10}$

$= \pi \left[\left(40 - \frac{40}{3}\right) - \left(-40 + \frac{40}{3}\right)\right]$

$= \frac{160}{3}\pi$ units^3

36) a)

$169y^2 = 676 - 25x^2$ so $y^2 = 4 - \frac{25}{169}x^2$

$\pi \int_{-5.2}^{5.2} y^2 \, \mathrm{d}x = \pi \int_{-5.2}^{5.2} 4 - \frac{25}{169}x^2 \, \mathrm{d}x$

$= \pi \left[4x - \frac{25}{507}x^3\right]_{-5.2}^{5.2}$

$= \pi \left[\left(\frac{104}{5} - \frac{104}{15}\right) - \left(-\frac{104}{5} + \frac{104}{15}\right)\right]$

$= \frac{416}{15}\pi$

$= 87.1$ mm^3 (3 SF)

36) b)

$25x^2 = 676 - 169y^2$ so $x^2 = \frac{676}{25} - \frac{169}{25}y^2$

$\pi \int_{-2}^{2} x^2 \, \mathrm{d}y = \pi \int_{-2}^{2} \frac{676}{25} - \frac{169}{25}y^2 \, \mathrm{d}y$

$= \pi \left[\frac{676}{25}y - \frac{169}{75}y^3\right]_{-2}^{2}$

$= \pi \left[\left(\frac{1352}{25} - \frac{1352}{75}\right) - \left(-\frac{1352}{25} + \frac{1352}{75}\right)\right]$

$= \frac{5408}{75}\pi$

$= 227$ mm^3 (3 SF)

36) c)

Rotating about the y-axis (part b) gives a shape that better matches the actual geometry of the chocolate — round and flattened, with circular top view and shorter height.